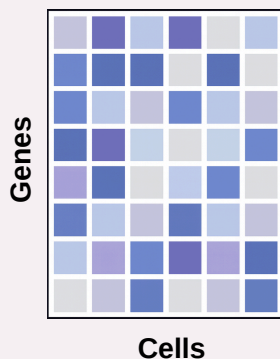
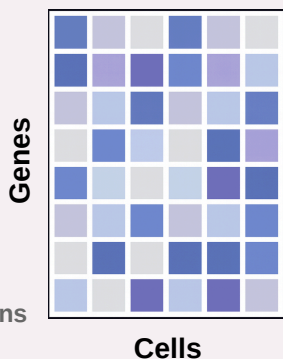


**Simulated
scRNA-seq data**
Controlled perturbations



Real scRNA-seq data
Public datasets

Reference methods	Proposed variants
PCA, PcaGrid, PcaHubert, PCP, K's tau	Winsor, Quad, Ball, Shell, LR

**Clustering
Performance**

**Subspace
Stability**

**Runtime
Scaling**

**Practical
guidance**